

DSA: In-order Traversal

COMPLETE TECHNICAL GUIDE • BINARY TREE ALGORITHMS

Introduction to In-order Traversal

In-order Traversal is a fundamental Depth First Search (DFS) algorithm used to visit all nodes in a binary tree. In this specific traversal strategy, we visit the left subtree first, then the root node, and finally the right subtree.

Key Characteristic: When performed on a Binary Search Tree (BST), In-order Traversal visits nodes in **ascending sorted order**.

The Recursive Logic

The "in" in In-order refers to the fact that the current node is visited in between the recursive calls to its children. The sequence follows a strict Left-Root-Right (L-R-R) pattern:

1. Traverse the left subtree by recursively calling the in-order function.
2. Visit the root node (typically print the data).
3. Traverse the right subtree by recursively calling the in-order function.

Python Implementation

The implementation is elegant and concise using recursion. Here is the standard Python code for performing an In-order Traversal on a binary tree node:

```
def inOrderTraversal(node):  
    if node is None:  
        return  
  
    # Step 1: Recur on left child  
    inOrderTraversal(node.left)  
  
    # Step 2: Visit the node  
    print(node.data, end=" ", "  
  
    # Step 3: Recur on right child  
    inOrderTraversal(node.right)
```

Step-by-Step Example Walkthrough

Consider a tree with Root **R**, Left Child **A** (with children C and D), and Right Child **B**.

- The function starts at R, but immediately calls `inOrder(A)`.
- A calls `inOrder(C)`. C has no left child, so it returns.
- '**C**' is printed.
- C has no right child, returns to A.
- '**A**' is printed.
- A calls `inOrder(D)`. '**D**' is printed.
- Finally, control returns to R. '**R**' is printed.

Resulting Output: **C, A, D, R, ...**

Complexity Analysis

Metric	Complexity
Time Complexity	$O(N)$ - where N is the number of nodes
Space Complexity	$O(H)$ - where H is the height of the tree (stack depth)

Applications of In-order Traversal

In-order traversal is not just a theoretical exercise; it has several practical uses in software engineering:

- **Sorting:** It is the primary method to retrieve elements from a BST in non-decreasing order.
- **Expression Trees:** Used to generate infix expressions from binary expression trees.
- **Validation:** Useful for checking if a binary tree is a valid BST by verifying if the resulting sequence is sorted.

Deep Dive: Recursion Stack in Traversal (Part 1)

To understand In-order traversal at 5-10 page depth, we must examine the call stack. When `inOrderTraversal(node.left)` is called, the current state of the function (local variables and the return address) is pushed onto the system stack.

Stack Trace insight: For a balanced tree of height H , the maximum memory used by the recursion is proportional to H . In the worst case (a skewed tree), this becomes $O(N)$.

This is why iterative implementations using an explicit stack are often preferred in production environments where tree depth might exceed the default recursion limit of the programming environment.

Algorithm Visualization Note

Imagine a tiny explorer walking along the "perimeter" of the binary tree. In an in-order traversal, the explorer only "marks" a node when they are passing it from directly underneath, between the left and right child clusters.

Deep Dive: Recursion Stack in Traversal (Part 2)

To understand In-order traversal at 5-10 page depth, we must examine the call stack. When `inOrderTraversal(node.left)` is called, the current state of the function (local variables and the return address) is pushed onto the system stack.

Stack Trace insight: For a balanced tree of height H , the maximum memory used by the recursion is proportional to H . In the worst case (a skewed tree), this becomes $O(N)$.

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Deep Dive: Recursion Stack in Traversal (Part 3)

To understand In-order traversal at 5-10 page depth, we must examine the call stack. When `inOrderTraversal(node.left)` is called, the current state of the function (local variables and the return address) is pushed onto the system stack.

Stack Trace insight: For a balanced tree of height H , the maximum memory used by the recursion is proportional to H . In the worst case (a skewed tree), this becomes $O(N)$.

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Deep Dive: Recursion Stack in Traversal (Part 4)

To understand In-order traversal at 5-10 page depth, we must examine the call stack. When `inOrderTraversal(node.left)` is called, the current state of the function (local variables and the return address) is pushed onto the system stack.

Stack Trace insight: For a balanced tree of height H , the maximum memory used by the recursion is proportional to H . In the worst case (a skewed tree), this becomes $O(N)$.

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